

Acoustic Form: Enhancing the Design Process for Assistive Earwear Jewelry via Material-Centric Computational Sound Analysis

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Figure 1: Assistive earwear jewelry with passive sound amplification.

Abstract

This work pioneers the application of computational acoustic simulation to assistive jewelry design for hearing enhancement. While acoustic modeling has been extensively used in traditional hearing aid development and architectural acoustics, its application to jewelry-scale assistive devices represents an unexplored design space. We present the first systematic methodology integrating acoustic simulation into the design process for assistive earwear jewelry, examining how material selection influences sound propagation in our originally designed Green Voice Smart Jewelry Device. Through computational analysis of four jewelry-grade metals—silver, titanium, platinum, and white gold—we demonstrate how acoustic simulation can inform both aesthetic and functional decisions in assistive jewelry design, establishing a new paradigm where computational acoustics guides the creation of hearing assistance devices that users want to wear.

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1 Introduction

Traditional hearing aids achieve technical performance but their medical appearance discourages use [Profita et al. 2018]. Decorative covers improve acceptance yet ignore acoustics. Conversely, hearing-aid engineering leverages sophisticated modeling but rarely considers jewelry form factors. Prior work on inclusive/assistive design [Pullin 2009] and finite-element analyses for auditory prostheses [Cheng et al. 2022] focuses on conventional devices, including feedback-path modeling [Sankowsky-Rothe et al. 2020]. We bridge this gap by introducing material-centric acoustic simulation to assistive jewelry. The Green Voice ear cuff incorporates internal channels intended to guide incident sound toward a capsule location while retaining an aesthetic, wearable form.

Computational analysis is particularly valuable for jewelry-scale devices, where physical prototyping in precious metals would be prohibitively expensive. Our simulation-first approach enables exploration of multiple materials—including platinum and gold alloys—without fabrication costs, allowing designers to make informed material decisions before committing to production.

2 Method

2.1 Framework

We used Treble’s hybrid acoustic engine, combining a wave-based discontinuous Galerkin solver for low frequencies with a ray-based (geometrical acoustics) solver at higher frequencies [Treble Technologies 2024]. This enabled full-spectrum modeling—capturing diffraction and interference—while maintaining computational efficiency. We note that Treble was initially developed for room-scale

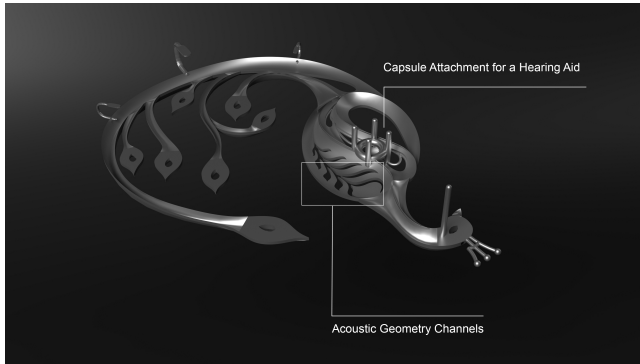


Figure 2: Green Voice ear cuff with internal channels guiding sound toward a receiver location.

acoustics, and adapting it to millimeter jewelry-scale presents some limitations.

2.2 Simulation Setup

The frequency sweep covered 250 Hz–8 kHz. A plane-wave source at 30 cm, 0° azimuth represented frontal conversation. Receivers were placed at the capsule and at reference points. Mesh resolution targeted 6–8 elements per wavelength at 8 kHz.

2.3 Materials

Material properties were represented by characteristic acoustic impedance

$$Z = \rho c,$$

where ρ is density and c is sound speed. The parameters used are listed in Table 1.

Table 1: Material properties used in simulation (Z in MRayl).

Material	ρ (kg/m ³)	c (m/s)	Z (MRayl)
Silver	10,500	3,650	38.3
Titanium	4,500	5,090	22.9
Platinum	21,450	2,680	57.5
White Gold (18K)	14,000–18,000	2,800–2,940	39.2–52.8

3 Results and Discussion

3.1 Material Behavior

All metals are highly mismatched to air ($Z_{\text{air}} \approx 415$ Rayl), so geometry dominates transmission. Within that constraint, simulations revealed characteristic tendencies: silver yielded balanced response; titanium produced smoother transmission and reduced phase irregularities near 2–4 kHz; platinum generated stronger boundary reflections and narrower effective collection; white gold spanned intermediate behaviors depending on alloy, suggesting tunability.

3.2 Frequency Trends

Divergence was minimal below 2 kHz where λ is large relative to features. Above 2 kHz, differences grew as wavelengths approached channel scales ($\lambda < 17$ cm), with response variations up to ~3 dB.

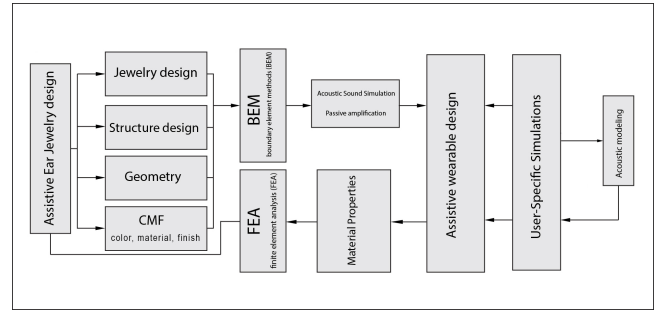


Figure 3: Workflow: geometry and material parameters → wave/BEM analysis → SPL maps and frequency responses for design feedback.

3.3 Implications

Designers could leverage material choice: silver/titanium for clarity, platinum for directional emphasis, white gold for customization. These insights extend beyond aesthetics, offering a rational basis for material selection in jewelry-scale assistance.

3.4 Limitations

Simulations omit head and pinna effects and boundary-layer physics at millimeter scales. Treble’s solvers are not specialized for such small structures, so results are indicative rather than predictive. Validation with FEM and vibrometry, along with psychoacoustic testing, is future work.

4 Conclusion

Material-aware acoustic simulation provides a new design-space lens for assistive jewelry. Though effects are subtle compared to traditional hearing aids, they are systematic and actionable at early stages. Future directions include refined solvers, mixed-material modeling, and psychoacoustic validation. The Green Voice ear cuff illustrates that assistive technology can be designed not only to function but to be worn proudly.

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